## H4 environmental-variable selection

**STEP 1**
**Inventory (230)**

**UCM**

AIR RH
AIR temp
HUMIDEX
MRT
PET
SND dBA
SUN Gh
UTCI
WIND AWS
AH
AIR T bot
AIR T mid
AIR T top
GPS speed kmh
HUMIDEX SR
IO flag
IR back
IR down
IR front
IR left
IR right
IR spot gnd
IR spot left
IR spot right
IR spot sky
IR up
MRT S
OPT
PET SR
SUN alt
SUN az
UTCI SR
WIND AWA
WIND TWD
WIND TWS
fp

**Atmo raw**

left pm1
left pm10
left pm2.5
right pm1
right pm10
right pm2.5
left humidity
left temperature
right humidity
right temperature

**Atmo average**

pm1 avg
pm10 avg
pm2.5 avg

**LYS1 raw**

LYS1 kelvin
LYS1 lux
LYS1 lux cat
LYS1 b'
LYS1 g'
LYS1 medi
LYS1 movement
LYS1 r'
LYS1 rgbb
LYS1 rgbg
LYS1 rgbir
LYS1 rgbr

**LYS2 raw**

LYS2 kelvin
LYS2 lux
LYS2 lux cat
LYS2 b'
LYS2 g'
LYS2 medi
LYS2 movement
LYS2 r'
LYS2 rgbb
LYS2 rgbg
LYS2 rgbir
LYS2 rgbr

**LYS average**

kelvin avg
lux avg
medi avg

**GVI imagery**

GVI 50m
GVI grass 50m
GVI phase mean
GVI plant 50m
GVI tree 50m
Grass phase mean
Plant phase mean
Tree phase mean

**SVF imagery**

SVF
Building %
Ceiling %
Combined Vegetation %
Grass %
Plant %
Sky %
Tree %
Wall %

**Sound**

band proportion 1000-4000 Hz
band proportion 125-1000 Hz
band proportion 20-125 Hz
band proportion 4000-8000 Hz
band proportion 8000-20000 Hz
category
dominant frequency (Hz)
mechanisms prop
mechanisms score
octave proportion 1000 Hz
octave proportion 125 Hz
octave proportion 2000 Hz
octave proportion 250 Hz
octave proportion 31.5 Hz
octave proportion 4000 Hz
octave proportion 500 Hz
octave proportion 63 Hz
octave proportion 8000 Hz
rural natural prop
rural natural score
spectral centroid (Hz)
spectral rolloff 85% (Hz)
unidentify prop
urban manmade negative prop
urban manmade negative score
urban manmade neutral prop
urban manmade neutral score
urban manmade positive prop
urban manmade positive score
vehicle prop
vehicle score
wind prop
wind score

**Solar**

Clearness Index
Clearness Index G0
Clearness Index Infer
SVF Aggregation Method
SVF Selection Method

**Empatica**

acc x
acc y
acc z
eda phasic
eda scl usiemens
eda tonic
heart rate
hrv fd hf
hrv fd hfnu
hrv fd lf
hrv fd lf hf ratio
hrv fd lfnu
hrv fd total power
hrv fd vlf
hrv td rmssd
hrv td sdnn
met
skin temperature
vector magnitude

**Eye-tracker**

blink duration s
distance from center
eyelid aperture avg
fixation duration s
fixation rate
gaze centrality
gaze dispersion
gaze velocity
pupil change rate
pupil diameter avg
saccade amplitude
saccade duration s
saccade peak velocity
stress composite

**Questionnaire**

current comfort 1 7
environmental comfort mean 1 5
pleasantness 1 5
prs11 being away mean 1 5
prs11 coherence mean 1 5
prs11 fascination mean 1 5
prs11 scope mean 1 5
prs11 total mean 1 5
space familiarity 1 5
stai6 mean 1 5
thermal comfort 1 5

**Metadata / QC**

Date
Date (Atmo/LYS)
Date (UCM)
Date (eye)
Datetime
GPS HDOP
GPS alt
GPS hdg
GPS lat
GPS lon
GPS speed valid sample count
GVI match dist m
HRV 5min available
HRV 5min max gap (s)
HRV 5min removed pct
HRV 5min segment end
HRV 5min segment start
HRV 5min valid rri count
HRV 5min valid span (s)
MAG hdg
ParticipantID
PhaseID
SVF Frames In Window
SVF Status
age
bike frequency
bmi category who
bmi kg m2
clothing accessories
clothing footwear
clothing height raw
clothing lower body
clothing skin tone
clothing upper body
clothing weight raw
commute modes
daily walking
end time s
gender
has children
height m
nationality
own bike for experiment
participant id
pct complete
phase start offset s
regular physical activity
residence type
residential area
response index
segment
site familiarity 1 10
vision correction
weight kg
years in brussels

### Parameters and decisions by source

**1. UCM**
- Retained: AIR RH, AIR temp, HUMIDEX, MRT, PET, SND dBA, SUN Gh, UTCI, WIND AWS.
- AH: Humidity representative: ucm_AIR_RH; AH is an alternative formulation
- AIR T bot, AIR T mid, AIR T top: Air-temperature representative: ucm_AIR_temp; tier channels not co-entered
- GPS speed kmh: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.
- HUMIDEX SR, MRT S, PET SR, UTCI SR: Derived alternate UCM formulations (_SR/_S suffix); use the corresponding unsuffixed UCM measure
- IO flag: Discrete indoor/outdoor setting flag excluded; experimental phase protocol already stratifies outdoor travel routes.
- IR back, IR down, IR front, IR left, IR right, IR spot gnd, IR spot left, IR spot right, IR spot sky, IR up: Directional IR radiometer channels (inputs used to compute ucm_MRT [CityFeel17]); measured solar exposure representative is ucm_SUN_Gh; not co-entered as separate radiation predictors
- OPT: Derived/redundant: ucm_OPT is operative temperature, derived from air temperature and MRT [CityFeel17]
- SUN alt, SUN az: Declared Step 2 exclusion: solar altitude and solar azimuth are outside the selected exposure set
- WIND AWA, WIND TWD: Derived wind direction/angle from ucm_WIND_AWS; wind speed representative is ucm_WIND_AWS. Circular direction coding is outside the current linear wind-speed specification.
- WIND TWS: Derived true-wind speed (device correction from AWS and related fields); wind representative: ucm_WIND_AWS
- fp: The definition, units and intended interpretation of fp have not been verified against source documentation; retain in the inventory but exclude from Step 2 analytical eligibility.

**2. Atmo raw**
- Inputs to retained averages: left pm1, left pm10, left pm2.5, right pm1, right pm10, right pm2.5. Highlighted as source measurements; not entered alongside their averages.
- left humidity, right humidity: Raw-source inventory only: UCM ucm_AIR_RH is the declared relative-humidity measure
- left temperature, right temperature: Raw-source inventory only: UCM ucm_AIR_temp is the declared ambient-temperature measure

**3. Atmo average**
- Retained: pm1 avg, pm10 avg, pm2.5 avg.

**4. LYS1 raw**
- Inputs to retained averages: LYS1 kelvin, LYS1 lux. Highlighted as source measurements; not entered alongside their averages.
- LYS1 lux cat: Derived illuminance category; continuous lux is available; no averaged replacement exists.
- LYS1 b', LYS1 g', LYS1 r', LYS1 rgbb, LYS1 rgbg, LYS1 rgbir, LYS1 rgbr: Exported RGB/IR or r'/g'/b' channel; no average or conversion to an independently calibrated H4 exposure is declared. Kept in the inventory, not falsely claimed to be represented by lux. Validate units/calibration and a spectral hypothesis before modeling.
- LYS1 medi: Source of lys_medi_avg; not co-entered with its own deterministic average. This is a study aggregation choice, not evidence that placement bias was eliminated. LYS1/LYS2 measure different wearer locations.
- LYS1 movement: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.

**5. LYS2 raw**
- Inputs to retained averages: LYS2 kelvin, LYS2 lux. Highlighted as source measurements; not entered alongside their averages.
- LYS2 lux cat: Derived illuminance category; continuous lux is available; no averaged replacement exists.
- LYS2 b', LYS2 g', LYS2 r', LYS2 rgbb, LYS2 rgbg, LYS2 rgbir, LYS2 rgbr: Exported RGB/IR or r'/g'/b' channel; no average or conversion to an independently calibrated H4 exposure is declared. Kept in the inventory, not falsely claimed to be represented by lux. Validate units/calibration and a spectral hypothesis before modeling.
- LYS2 medi: Source of lys_medi_avg; not co-entered with its own deterministic average. This is a study aggregation choice, not evidence that placement bias was eliminated. LYS1/LYS2 measure different wearer locations.
- LYS2 movement: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.

**6. LYS average**
- Retained: kelvin avg, lux avg.
- medi avg: Derived metric estimated from LYS RGB channels, not measured spectrally; retained light measures lys_lux_avg and lys_kelvin_avg

**7. GVI imagery**
- Retained: GVI 50m.
- GVI grass 50m, GVI phase mean, GVI plant 50m, GVI tree 50m, Grass phase mean, Plant phase mean, Tree phase mean: Vegetation representative: gvi_50m; alternative scales/components not co-entered

**8. SVF imagery**
- Retained: SVF.
- Building %, Ceiling %, Combined Vegetation %, Grass %, Plant %, Sky %, Tree %, Wall %: Sky openness represented by SVF; scene percentages are separate secondary constructs, not equivalent exposures

**9. Sound**
- band proportion 1000-4000 Hz, band proportion 125-1000 Hz, band proportion 20-125 Hz, band proportion 4000-8000 Hz, band proportion 8000-20000 Hz, dominant frequency (Hz), octave proportion 1000 Hz, octave proportion 125 Hz, octave proportion 2000 Hz, octave proportion 250 Hz, octave proportion 31.5 Hz, octave proportion 4000 Hz, octave proportion 500 Hz, octave proportion 63 Hz, octave proportion 8000 Hz, spectral centroid (Hz), spectral rolloff 85% (Hz): Omitted by the current specification. Scientific justification remains unresolved; dBA does not replace frequency information. No separate frequency analysis is implemented.
- category, mechanisms prop, mechanisms score, rural natural prop, rural natural score, unidentify prop, urban manmade negative prop, urban manmade negative score, urban manmade neutral prop, urban manmade neutral score, urban manmade positive prop, urban manmade positive score, vehicle prop, vehicle score, wind prop, wind score: Omitted by the current specification. Sound type is distinct from dBA; no separate sound-type analysis is implemented.

**10. Solar**
- Clearness Index: Derived clearness index (kt = Gh/G0); measured solar exposure is ucm_SUN_Gh [ClearCI p.3 eq.2]
- Clearness Index G0: Calculated extraterrestrial irradiance G0 (date/time/location only), not a measured or personal exposure; denominator of Clearness_Index (kt = Gh/G0) [ClearCI p.3 eq.2]; measured solar exposure is ucm_SUN_Gh.
- Clearness Index Infer: Derived categorical clearness inference from Clearness_Index; not a continuous numeric exposure; no valid continuous-effect parameterization is declared. The original field remains in the input. Measured solar exposure is ucm_SUN_Gh
- SVF Aggregation Method, SVF Selection Method: Not a continuous numeric exposure; no valid continuous-effect parameterization is declared. The original field remains in the input.

**11. Empatica**
- acc x, acc y, acc z, met, vector magnitude: Movement variable retained in the complete inventory; it is not an H4 exposure predictor. No LYS movement average is declared.
- eda phasic, eda tonic, heart rate, hrv fd hf, hrv fd hfnu, hrv fd lf, hrv fd lf hf ratio, hrv fd lfnu, hrv fd total power, hrv fd vlf, hrv td rmssd, hrv td sdnn, skin temperature: Used as a physiological or ocular response, not an environmental predictor.
- eda scl usiemens: Signal quality, missingness verification, or HRV window integrity metric.

**12. Eye-tracker**
- blink duration s, distance from center, eyelid aperture avg, fixation duration s, fixation rate, gaze centrality, gaze dispersion, gaze velocity, pupil change rate, pupil diameter avg, saccade amplitude, saccade duration s, saccade peak velocity: Used as a physiological or ocular response, not an environmental predictor.
- stress composite: Signal quality, missingness verification, or HRV window integrity metric.

**13. Questionnaire**
- current comfort 1 7, environmental comfort mean 1 5, pleasantness 1 5, prs11 being away mean 1 5, prs11 coherence mean 1 5, prs11 fascination mean 1 5, prs11 scope mean 1 5, prs11 total mean 1 5, space familiarity 1 5, stai6 mean 1 5, thermal comfort 1 5: Signal quality, missingness verification, or HRV window integrity metric.

**14. Metadata / QC**
- Date, Date (Atmo/LYS), Date (UCM), Date (eye), Datetime, GPS HDOP, GPS alt, GPS hdg, GPS lat, GPS lon, GPS speed valid sample count, GVI match dist m, MAG hdg, ParticipantID, PhaseID, pct complete: Spatiotemporal tracking, GPS positioning, or phase grouping metadata.
- HRV 5min available, HRV 5min max gap (s), HRV 5min removed pct, HRV 5min segment end, HRV 5min segment start, HRV 5min valid rri count, HRV 5min valid span (s): Signal quality, missingness verification, or HRV window integrity metric.
- SVF Frames In Window, SVF Status: SVF calibration, segmentation, run-integrity, telemetry, or manual-validation QC; not an exposure predictor.
- age, bike frequency, bmi category who, bmi kg m2, clothing accessories, clothing footwear, clothing height raw, clothing lower body, clothing skin tone, clothing upper body, clothing weight raw, commute modes, daily walking, gender, has children, height m, nationality, own bike for experiment, participant id, regular physical activity, residence type, residential area, response index, site familiarity 1 10, vision correction, weight kg, years in brussels: Baseline participant demographic characteristics or clothing insulation metadata.
- end time s, phase start offset s, segment: YAMNet/spectral processing provenance, completeness, timing, or frequency-coverage QC; not an exposure predictor.

---

**STEP 2**
**Domain Grouping (16)**

| Thermal | Index | Acoustic | Radiation | Pollution | Light | Greenery |
|---|---|---|---|---|---|---|
| AIR temp | HUMIDEX | SND dBA | MRT | pm1 avg | lux avg | GVI 50m |
| AIR RH | PET | | SUN Gh | pm2.5 avg | kelvin avg | |
| WIND AWS | UTCI | | SVF | pm10 avg | | |

**STEP 3**
**Equations + design VIF**
**(7 equations)**

| | |
|---|---|
| Eq 1 (Primary Model): UTCI * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM2.5 + lux + kelvin (light) + GVI (greenery) | max VIF = 3.07 |
| Eq 2 (Sound sensitivity): Eq 1 + five family-wise PCA-screened sound mains (traffic, low-freq band, wind, spectral centroid, mechanisms) | max VIF = 3.07 |
| Eq 3 (Heat sensitivity — air temp, RH, wind): air temp * Sound + RH * Sound + wind * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM2.5 + lux + kelvin (light) + GVI (greenery) | max VIF = 2.55 |
| Eq 4 (Heat sensitivity — HUMIDEX): HUMIDEX * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM2.5 + lux + kelvin (light) + GVI (greenery) | max VIF = 2.50 |
| Eq 5 (Heat sensitivity — PET): PET * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM2.5 + lux + kelvin (light) + GVI (greenery) | max VIF = 3.02 |
| Eq 6 (PM size sensitivity — PM1): UTCI * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM1 + lux + kelvin (light) + GVI (greenery) | max VIF = 3.06 |
| Eq 7 (PM size sensitivity — PM10): UTCI * Sound + MRT * Sound + solar irradiance + SVF (sky view) + PM10 + lux + kelvin (light) + GVI (greenery) | max VIF = 3.07 |